

HDQC-25

Aim:

To write a menu-driven C program using a switch statement, nested loops, and separate parameter-less functions to display different decreasing patterns based on the user's choice.

Task1: Algorithm:

Step 1: Start.

Step 2: Declare choice as an integer.

Step 3: Define four parameterless functions with void return type:

- pattern1() – Decreasing right-angled triangle
- pattern2() – Decreasing natural-number pattern
- pattern3() – Decreasing repeated-number pattern
- pattern4() – Reverse-number triangle

Step 4: Display the menu:

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Step 5: Input the user's choice.

Step 6: Use a switch statement:

- If choice = 1, call pattern1().
- If choice = 2, call pattern2().
- If choice = 3, call pattern3().
- If choice = 4, call pattern4().
- If choice = 5, terminate the program.
- Otherwise, display "Invalid choice!".

Step 7: In each pattern function:

1. Input number of rows n.
2. Check whether n is positive.
3. Use nested for loops to generate the required pattern.

Step 8: Repeat the menu until the user selects Exit.

Step 9: Stop.

Task2: Test case table:

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Choice = 1, n = 4	* * * * *	Same as expected	Pass
2	Choice = 2, n = 4	1 2 3 4 1 2 3 1 2 1	Same as expected	Pass
3	Choice = 3, n = 4	4 4 4 4 3 3 3 2 2 1	Same as expected	Pass
4	Choice = 4, n = 4	4 3 2 1 3 2 1 2 1 1	Same as expected	Pass
5	Choice = 2, n = 0	Invalid number of rows!	Invalid number of rows!	Pass
6	Choice = 3, n = -2	Invalid number of rows!	Invalid number of rows!	Pass
7	Choice = 7	Invalid choice!	Invalid choice!	Pass
8	Choice = 5	Exiting program...	Exiting program...	Pass

Task3: C program:

```
#include <stdio.h>
void pattern1();
void pattern2();
void pattern3();
void pattern4();
int main()
{
    int choice;
    do
    {
        printf("\n----- DECREASING PATTERNS ----- \n");
        printf("1. Decreasing right-angled triangle\n");
        printf("2. Decreasing natural-number pattern\n");
        printf("3. Decreasing repeated-number pattern\n");
        printf("4. Reverse-number triangle\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
```

```

scanf("%d", &choice);
switch(choice)
{
    case 1:
        pattern1();
        break;
    case 2:
        pattern2();
        break;
    case 3:
        pattern3();
        break;
    case 4:
        pattern4();
        break;
    case 5:
        printf("Exiting program...\n");
        break;
    default:
        printf("Invalid choice!\n");
}
} while(choice != 5);
return 0;
}
/* Pattern 1 */
void pattern1()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if(n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }
    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= i; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
/* Pattern 2 */
void pattern2()
{

```

```

int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if(n <= 0)
{
    printf("Invalid number of rows!\n");
    return;
}
for(i = n; i >= 1; i--)
{
    for(j = 1; j <= i; j++)
    {
        printf("%d ", j);
    }
    printf("\n");
}
}

/* Pattern 3 */
void pattern3()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if(n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }
    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", i);
        }
        printf("\n");
    }
}

/* Pattern 4 */
void pattern4()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    if(n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }

```

```

    }
    for(i = n; i >= 1; i--)
    {
        for(j = i; j >= 1; j--)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

```

Task4: Output:

```

(kali@kali)~/hdoc25.c
$ ./a.out

—DECREASING PATTERNS—
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*
*

—DECREASING PATTERNS—
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 2
Enter number of rows: 4
1234
123
12
1

—DECREASING PATTERNS—
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 3
Enter number of rows: 4
4444
333
22
1

```

```
(kali@kali) ~$ ./a.out
```

——DECREASING PATTERNS——

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 4

Enter number of rows: 4

4321

321

21

1

——DECREASING PATTERNS——

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 2

Enter number of rows: 0

Invalid number of rows!

——DECREASING PATTERNS——

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 3

Enter number of rows: -2

Invalid number of rows!

——DECREASING PATTERNS——

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 7

Invalid choice!

——DECREASING PATTERNS——

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit

Enter your choice: 5

Exiting program...